Seung-Yoon Back

Ph.D. · Atmospheric & Climate Dynamics

Postdoctoral Researcher, Department of the Geophysical Sciences, The University of Chicago
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EDUCATION

Ph.D., Earth and Environmental Sciences Sep 2020 – Aug 2025
School of Earth and Environmental Sciences, Seoul National University
Thesis: Modulation of MJO Characteristics by the Tropospheric Moisture Distribution and Stratospheric Wind
Advisor: Prof. Seok-Woo Son · Co-advisor: Prof. Daehyun Kim

M.Sc., Earth and Environmental Sciences Mar 2018 – Feb 2020
School of Earth and Environmental Sciences, Seoul National University
Thesis: QBO Influence on MJO Convection in a Numerical Weather Prediction Model
Advisor: Prof. Seok-Woo Son

B.Sc., Earth Science Education Mar 2014 – Feb 2018
Department of Earth Science Education, Seoul National University, Korea

PROFESSIONAL APPOINTMENTS

Postdoctoral Researcher Sep 2025 – Present
Department of the Geophysical Sciences, The University of Chicago

Researcher Sep 2023 – Aug 2025
Research Institute of Basic Sciences, Seoul National University

HONORS & AWARDS

- Best Doctoral Thesis Award, Korean Meteorological Society (Nov 2025)
- Best Doctoral Thesis Award, College of Natural Sciences, Seoul National University (Aug 2025)
- Best Student Poster Presentation Award, Korean Meteorological Society (Apr 2022)
- Outstanding Paper Award, Korean Society of Climate Change Research (Sep 2018) — *Classification of Heat Wave Events in Seoul Using a Self-Organizing Map*

RESEARCH INTERESTS

Atmospheric and climate dynamics, with emphasis on **QBO dynamics**, **MJO dynamics**, and **stratosphere–troposphere coupling**.

PUBLICATIONS

** corresponding author*

Peer-Reviewed Journal Articles

1. Elsbury, D.*, F. Serva, J. M. Caron, **S.-Y. Back**, C. Orbe, …, S.-W. Son, and Coauthors, 2026: QBOi El Niño–Southern Oscillation Experiments: Assessing Relationships between ENSO, MJO, and QBO. *Weather and Climate Dynamics*, 7, 317–339.
2. **Back, S.-Y.**, D. Kim*, S.-W. Son*, and D. Kang, 2025: Interseasonal Difference in MJO Propagation during the Extended Boreal Winter: Observations and Climate Model Simulations. *Journal of Climate*, 38, 6693–6708.
3. **Back, S.-Y.**, D. Kim*, and S.-W. Son*, 2024: MJO Diversity in CMIP6 Models. *Journal of Climate*, 37, 4835–4850.

4. Zhao, Y., S.-W. Son*, and **S.-Y. Back**, 2023: The Critical Role of the Upper-Level Synoptic Disturbance on the China Henan "21.7" Extreme Precipitation Event. Scientific Online Letters on the Atmosphere, 19, 42–49.
5. **Back, S.-Y.**, J.-Y. Han, and S.-W. Son*, 2020: Modeling Evidence of QBO–MJO Connection: A Case Study. *Geophysical Research Letters*, 47, e2020GL089480.

Under Review / In Preparation

6. Huang, K.*, C.-H. Park*, **S.-Y. Back**, …, and Coauthors: Missing QBO–MJO Connection in the QBOi Phase 2 Climate Models. *Submitted*.
7. **Back, S.-Y.***, T. A. Shaw, and D. Yang: Sensitivity of a QBO-like Oscillation to Convective Self-Aggregation in a Radiative–Convective Equilibrium. *In preparation*.
8. **Back, S.-Y.**, S.-W. Son*, and J.-H. Kim: Mechanism-Denial Study on the QBO–MJO Connection in a Regional Model. *In preparation*.

Domestic Journal Articles (in Korean, with English abstract)

9. Kim, H., Y. Kwon*, **S.-Y. Back**, J. Hwang, S.-W. Son, H. Park, and E.-J. Cha, 2024: Evaluation of Short-Term Prediction Skill of East Asian Summer Atmospheric Rivers. *Atmosphere*, 34, 83–95.
10. Kim, G., **S.-Y. Back**, Y. Kwon, and S.-W. Son*, 2023: Comparison of Atmospheric River Detection Algorithms in East Asia. *Atmosphere*, 33, 399–411.
11. Oh, J., S.-W. Son*, and **S.-Y. Back**, 2022: Influence of UTLS Ozone on the QBO–MJO Connection: A Case Study Using the GloSea5 Model. *Atmosphere*, 32, 223–233.
12. Kwon, Y., C. Park, **S.-Y. Back***, S.-W. Son, J. Kim, and E.-J. Cha, 2022: Influence of Atmospheric Rivers on Regional Precipitation in South Korea. *Atmosphere*, 32, 1–14.
13. **Back, S.-Y.**, S.-W. Kim, M.-I. Jung, J.-W. Roh, and S.-W. Son*, 2018: Classification of Heat Wave Events in Seoul Using a Self-Organizing Map. *Journal of Climate Change Research*, 9, 209–221.

SELECTED PRESENTATIONS

MJO Diversity across Timescales: Observations and Model Simulations — Sep 2025
Seoul National University, Seoul, Korea [Oral]

Interseasonal Difference in MJO Propagation during the Extended Boreal Winter — Jul 2025
BACO25, Pusan, Korea [Oral]

Modulation of MJO Characteristics by the Tropospheric Moisture Distribution — Jul 2025
Korea Institute of Science and Technology, Seoul, Korea [Oral, invited]

QBO–MJO Connection in a Hierarchy of Models — Oct 2024
Pukyong National University, Pusan, Korea [Oral, invited]

MJO Diversity in CMIP6 Models — Jun 2024
Asia-Oceania Geosciences Society, Pyeongchang, Korea [Oral]

Seasonality of MJO Diversity — Jun 2024
Asia-Oceania Geosciences Society, Pyeongchang, Korea [Poster]

Subseasonal Prediction of East Asian Atmospheric Rivers — Jun 2024
Asia-Oceania Geosciences Society, Pyeongchang, Korea [Poster]

Evaluation of Short-Term Prediction Skill of East Asian Summer Atmospheric Rivers — Apr 2023
Korean Meteorological Society, Pusan, Korea [Poster]

Comparison of Atmospheric River Detection Algorithms in East Asia — Apr 2023
Korean Meteorological Society, Pusan, Korea [Poster]

Synoptic Analysis and Numerical Experiments of Heavy Rainfall Events in Seoul, August 2022 — Apr 2023
Korean Meteorological Society, Pusan, Korea [Poster]

MJO Diversity in CMIP6 Models — Apr 2023
Korean Meteorological Society, Pusan, Korea [Oral]

The QBO and ENSO Influence on MJO Diversity — Mar 2023
QBOi Workshop, Oxford, UK [Poster]

MJO Diversity in CMIP6 Models — Dec 2022
AGU Fall Meeting, Chicago, USA [Poster]

MJO Diversity in CMIP6 Models — May 2022
EGU General Assembly, Vienna, Austria [Oral, virtual]

MJO Diversity in CMIP6 Models — Apr 2022
Korean Meteorological Society, Pusan, Korea [Poster]

Modeling Evidence of QBO–MJO Connection: A Case Study — Aug 2021
Asia-Oceania Geosciences Society, Singapore [Poster, virtual]

Modeling Evidence of QBO–MJO Connection: A Case Study — Oct 2020
Korean Meteorological Society, Gyeongju, Korea [Oral, virtual]

Numerical Simulation of QBO–MJO Connection — Feb 2020
National Institute of Meteorological Sciences, Jeju, Korea [Oral]

Classification of Heat Wave Events in Seoul Using a Self-Organizing Map — Feb 2020
National Institute of Meteorological Sciences, Jeju, Korea [Oral, invited]

Numerical Simulation of QBO–MJO Connection — Oct 2019
Korean Meteorological Society, Gyeongju, Korea [Poster]

QBO–MJO Coupling in a Numerical Weather Prediction Model — May 2019
Korean Meteorological Society, Daegu, Korea [Poster]

Classification of Korean Heatwaves: Characteristics and Related Synoptic Patterns — Oct 2018
Korean Meteorological Society, Jeju, Korea [Poster]

Classification of Heat Wave Events in Seoul Using a Self-Organizing Map — Apr 2018
Korean Meteorological Society, Seoul, Korea [Poster]

COLLABORATIVE PROJECTS

- **QUOCA** — Quasi-Biennial Oscillation and Ozone Chemistry Interactions in the Atmosphere (APARC), 2025–present
- **QBOi** — QBO–ENSO Experiment (APARC), 2022–present
- **Visiting collaborator**, Center for Advanced Data Assimilation and Predictability Techniques, The Pennsylvania State University (Prof. Fuqing Zhang), Oct 2018

PROFESSIONAL SERVICE

Peer Review

Nature Communications, Journal of Climate, Journal of Geophysical Research: Atmospheres, Climate Dynamics

Teaching (Teaching Assistant, Seoul National University)

- Large-Scale Dynamics of the Atmosphere (Spring 2022)
- Atmospheric Science Laboratory (Fall 2020)
- Weather Forecasting and Laboratory (Spring 2018)
- Student Teacher, Seoul National University Middle School (May 2017)